

IONIC BASIS OF CARDIAC ACTION POTENTIAL

Headings

▼ IONIC BASIS OF CARDIAC ACTION POTENTIAL

Headings

▼ PHASES AND IONIC MECHANISMS

▼ DETAILED BREAKDOWN OF PHASES

1. Initial Depolarization (Phase 0)
2. Initial Repolarization (Phase 1)
3. Final Depolarization / Plateau (Phase 2)
4. Final Repolarization (Phase 3)

▼ RESTORATION OF RESTING MEMBRANE POTENTIAL (Phase 4)

RESTORATION OF RESTING MEMBRANE POTENTIAL (PHASE 4)
SPREAD OF ACTION POTENTIAL

▼ PACEMAKER OF THE HEART

SINOATRIAL (SA) NODE

RHYTHMICITY / RATE OF IMPULSE DISCHARGE

- **Resting Membrane Potential (RMP) :—**
 - **Single Cardiac Muscle Fiber** : -85 to -95 mV
 - **Sinoatrial Node (SA Node)** : -55 to -60 mV
 - **Purkinje Fibers** : -90 to -100 mV
- **Action Potential Duration** : $250 - 350$ ms ($0.25 - 0.35$ sec).
- **4 Phases of Action Potential :—**
 - Phase 0 : Initial Depolarization
 - Phase 1 : Initial Rapid Repolarization
 - Phase 2 : Plateau / Final Depolarization (Prolonged state)
 - Phase 3 : Final Repolarization

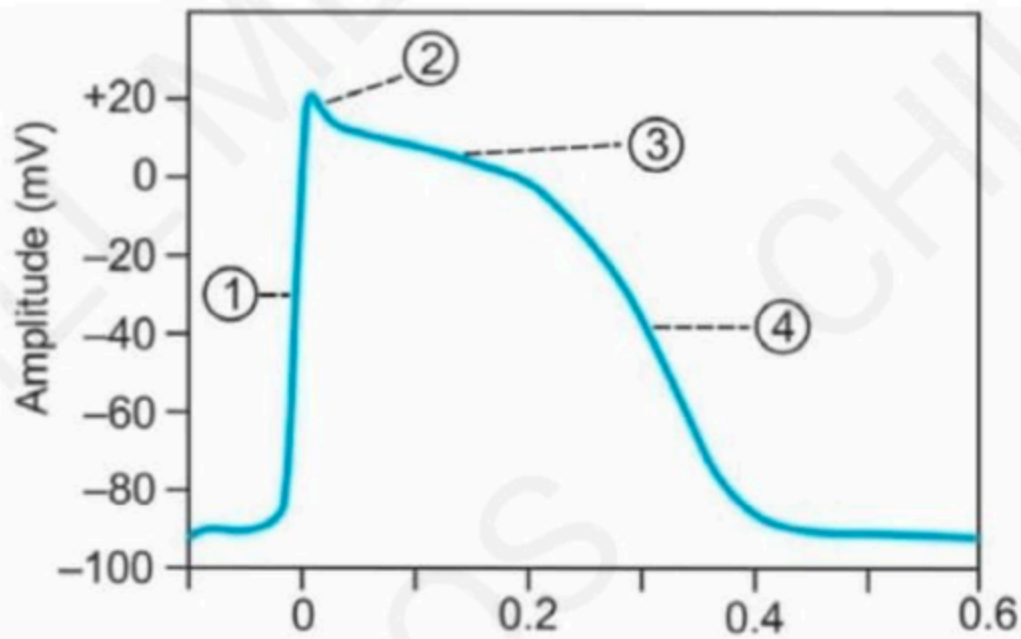


FIGURE 90.1: Action potential in ventricular muscle. 1 = Depolarization, 2 = Initial rapid repolarization, 3 = Plateau, 4 = Final repolarization.

PHASES AND IONIC MECHANISMS

**Phase 0: Initial
Depolarization**

Rapid opening of Fast Na^+
channels
Rapid Na^+ influx
(Overshoot to +20 mV)



**Phase 1: Initial
Repolarization**

Closure of Fast Na^+
channels
Transient K^+ efflux
(Produces a notch)



**Phase 2: Plateau (Final
Depolarization)**

Slow opening of L-type
 Ca^{2+} channels
 Ca^{2+} & slow Na^+ influx
balances K^+ efflux



**Phase 3: Final
Repolarization**

Closure of Ca^{2+} channels
Rapid K^+ efflux restores
negativity



**Phase 4: Restoration of
RMP**

Na^+ - K^+ ATPase pump & Na^+ -
 Ca^{2+} exchanger

Restores ionic equilibrium

DETAILED BREAKDOWN OF PHASES

1. Initial Depolarization (Phase 0)

- **Mechanism** : Caused by the **rapid opening of fast Na^+ channels**, resulting in a rapid influx of Na^+ ions into the cell (similar to skeletal muscle fibers).
- **Duration** : Lasts for about 2 msec (0.002 sec).
- **Amplitude** : Membrane potential rises to an amplitude of about +20 mV.

2. Initial Repolarization (Phase 1)

- **Mechanism** :
 - Due to the **transient (short duration) opening of K^+ channels** causing an efflux of a small quantity of K^+ ions.
 - The fast Na^+ channels **close suddenly**, while slow Na^+ channels open, resulting in a slow influx of a low quantity of Na^+ ions.
- **Duration** : Lasts for 2 msec.
- **Morphology** : Represented as a distinct **notch** on the action potential curve.

3. Final Depolarization / Plateau (Phase 2)

- **Mechanism** :

Slow opening of Ca^{2+} channels (remain open for a longer period)



Influx of large number of Ca^{2+} ions



Slow influx of Na^+ ions (through already opened slow Na^+ channels)



Prolonged depolarization (Ca^{2+} & Na^+ ions maintain positivity inside)

- **Duration :**
 - 200 msec in atrial muscle fibers.
 - 300 msec in ventricular muscle fibers.
- **Functional Consequence :** Contraction time is **5 to 15 times longer** in cardiac muscle than in skeletal muscle.

4. Final Repolarization (Phase 3)

- **Mechanism :**
 - Due to marked **efflux of K^+ ions**.
 - The number of K^+ ions moving out exceeds the number of Ca^{2+} ions moving in, restoring intracellular negativity (**final repolarization**).
- **Duration :** A slow process that lasts for 50 — 80 msec.

RESTORATION OF RESTING MEMBRANE POTENTIAL (Phase 4)

Activation of $\text{Na}^+ - \text{K}^+$ pump $\longrightarrow$ All Na^+ ions move out & K^+ ions move into the cell



Activation of $\text{Na}^+ - \text{Ca}^{2+}$ pump $\longrightarrow$ Excess of Ca^{2+} ions move out

RESTORATION OF RESTING MEMBRANE POTENTIAL (PHASE 4)

- **Na⁺-K⁺ Pump (ATPase)** : Actively pumps Na⁺ ions out of the cell and moves K⁺ ions back into the cell (3 Na⁺ out/2 K⁺ in).
- **Na⁺-Ca²⁺ Exchanger** : Extrudes excess intracellular Ca²⁺ ions out of the fiber.

SPREAD OF ACTION POTENTIAL

Action potential spreads very rapidly across cardiac muscle



Due to presence of **Gap Junctions** between intercalated discs of cardiac muscle fibers



Gap junctions are highly permeable, allowing free bidirectional movement of ions



Facilitates rapid propagation from fiber to fiber (**Functional Syncytium**)

PACEMAKER OF THE HEART

- **Definition** : Structure of the heart that spontaneously generates electrical impulses responsible for the heartbeat.
- **Cell Type** : Formed by specialized, auto-rhythmic cells called **P-cells (Pacemaker cells)**.
- **Historical Discovery** : **Sir Thomas Lewis** identified and named the SA node as the primary *pacemaker of the heart*.

SINOATRIAL (SA) NODE

- **Anatomy** : A small strip of specialized modified cardiac muscle situated in the **superior part of the lateral wall of the right atrium**, immediately below the opening of the superior vena cava (SVC).

- **Histology** : SA nodal fibers **lack contractile elements (myofibrils)**.
- **Conduction** : Fibers are continuous with ordinary atrial muscle fibers, allowing impulses to spread rapidly throughout both atria.

RHYTHMICITY / RATE OF IMPULSE DISCHARGE

The atrioventricular (AV) node, atria, and Purkinje system can also generate impulses and function as potential/latent pacemakers:

Cardiac Structure / Pacemaker Focus	Rate of Impulse Production
1. SA Node	70 – 80 / min
2. AV Node	40 – 60 / min
3. Atrial Muscle	40 – 60 / min
4. Ventricular Muscle	20 – 40 / min
5. Purkinje Fibers	35 – 40 / min